

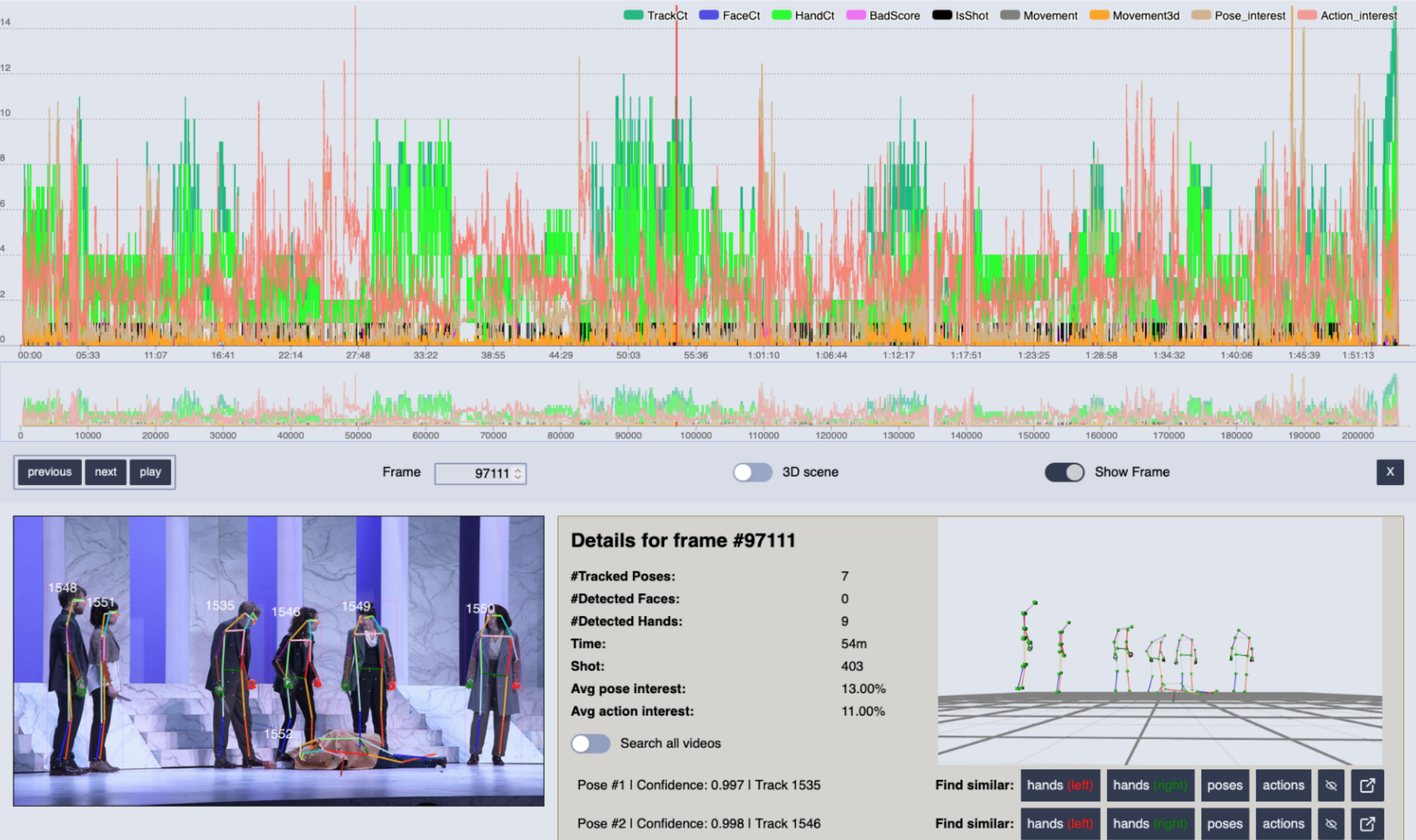
Stylistic Analyses of Human Pose in Theatrical Performances

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**Approach:** we explore how classifiers given pose, action, and inferred motion and distance features can assign each of 31 full-length performances to the correct director, from among three “auteurs” (Jones, Castellucci, Warlikowski).

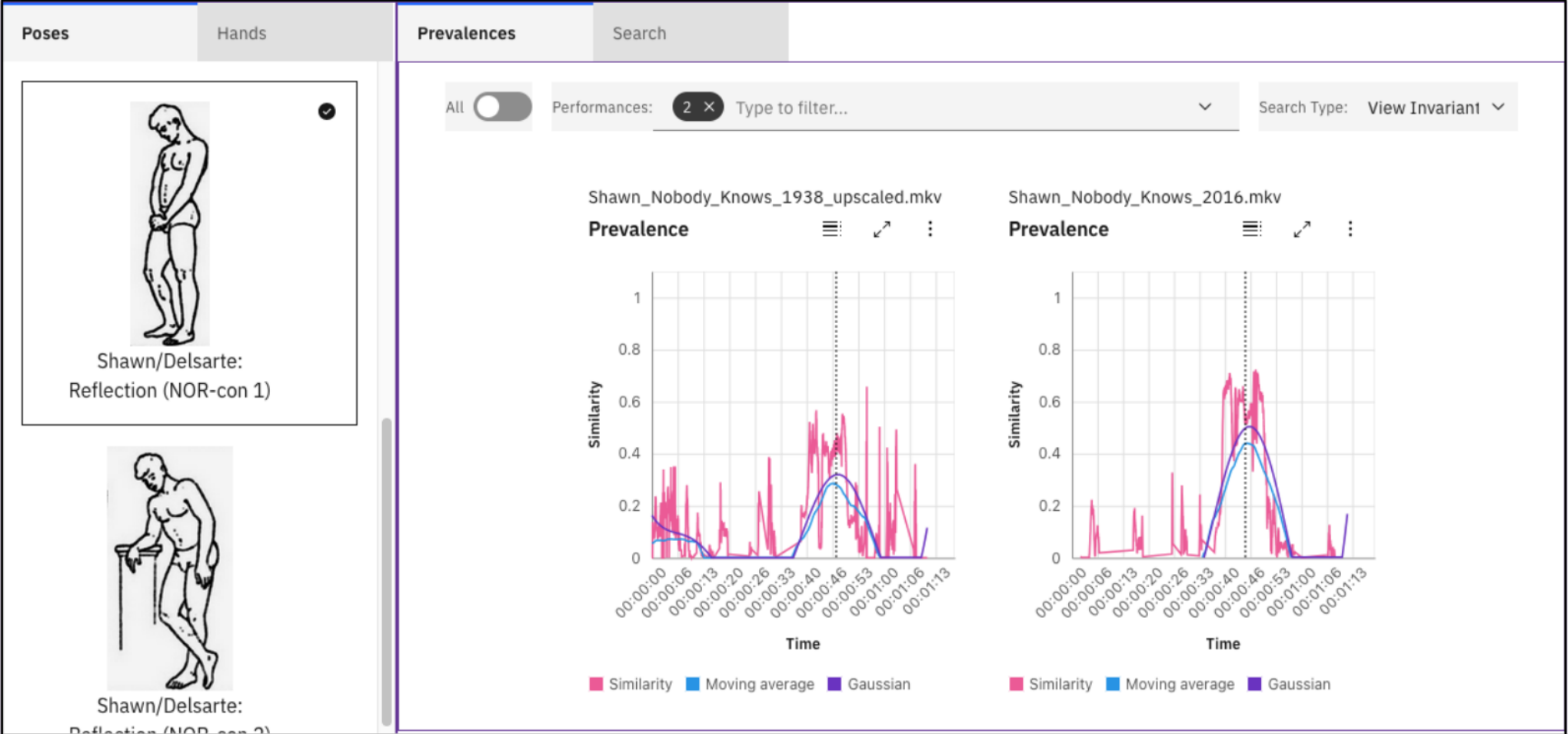
**Analysis:** how do the stylometric findings relate to conventional wisdom regarding directors’ signature choreographic styles? Can theories of acting, dance and oratory from earlier eras help to interpret the results?



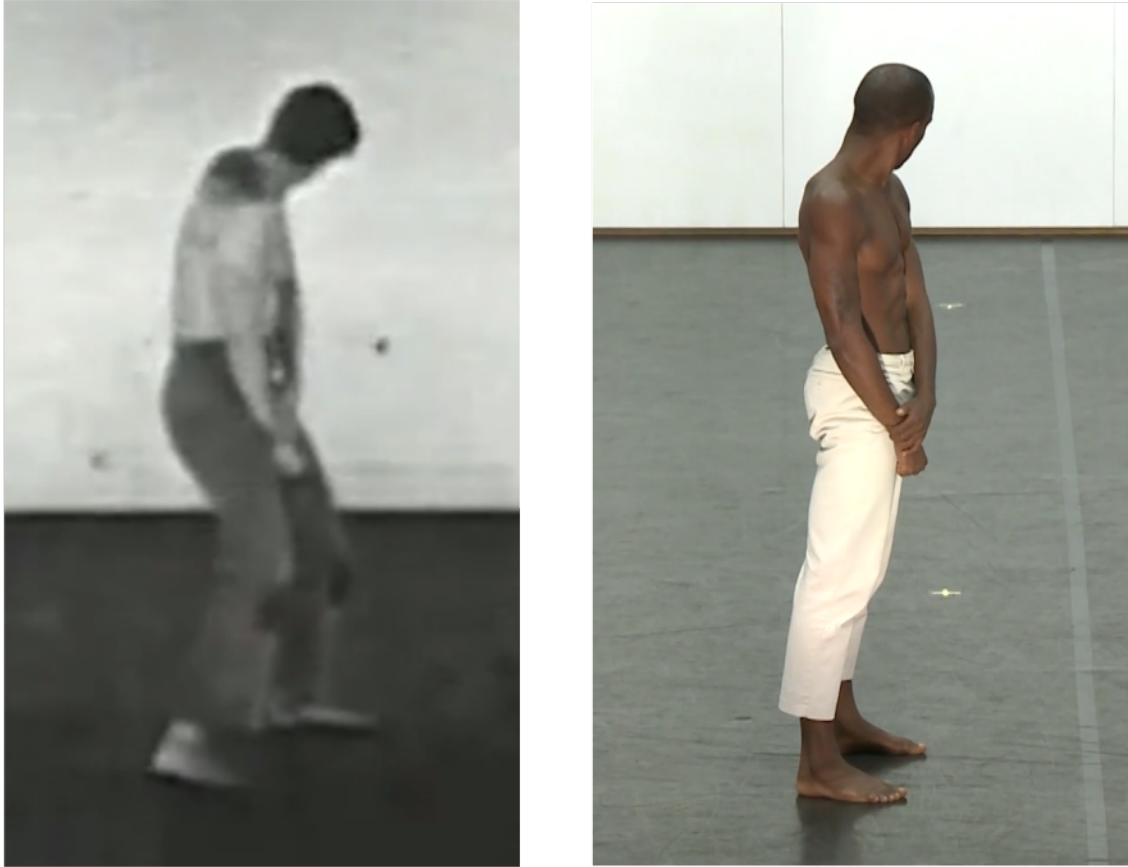
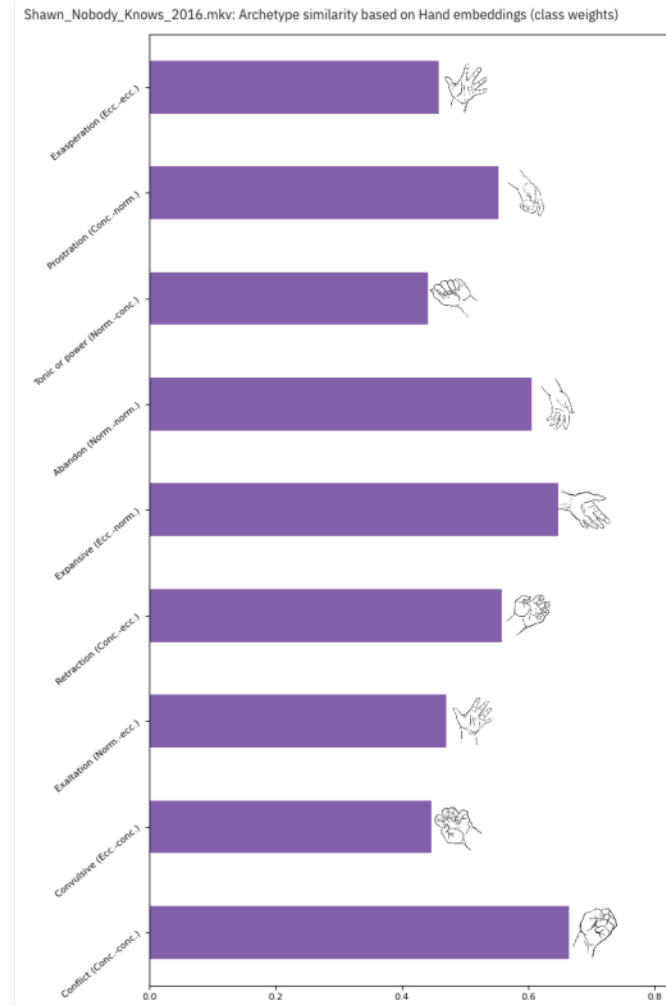
The timeline view of the MIME (Machine Intelligence for Motion Exegesis) application, displaying pose and motion data for a performance, with details and 3D body armatures for a selected frame.

|                       | Motion & distance  | Pose embeddings    | 3D global coordinates | Action embeddings  |
|-----------------------|--------------------|--------------------|-----------------------|--------------------|
| LOO cosine similarity | 51.61% (16/31)     | 77.42% (24/31)     | 74.19% (23/31)        | 80.65% (25/31)     |
| 10-fold Random Forest | 57.5% stdev: 37.4% | 66.6% stdev: 33.7% | 66.6% stdev: 33.7%    | 68.3% stdev: 31.3% |
| 10-fold Gaussian NB   | 76.7% stdev: 26.0% | 75.3% stdev: 25.5% | 72.1% stdev: 25.3%    | 76.9% stdev: 24.5% |

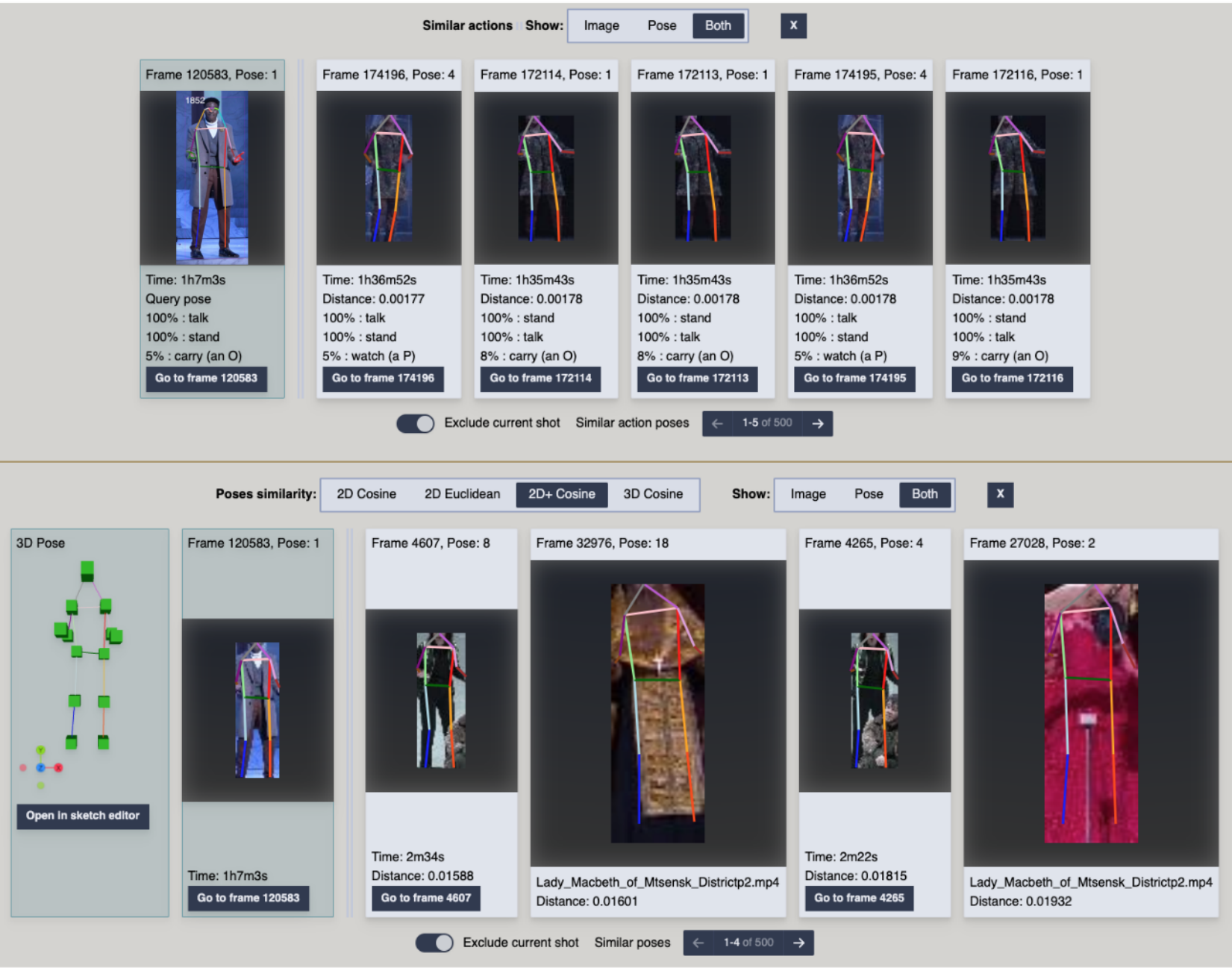
Accuracies of the specified classification approaches for the 31 performances and three directors, given different pose-related features; statistics from the Random Forest and Gaussian Naive Bayes (NB)-based classifiers reflect 10-fold cross validation.



Pose prevalence time-series plots based on cosine similarity of the detected poses to the “Reflection” pose archetype from the system of François Delsarte (1811-1871), as illustrated by American modern dance luminary Ted Shawn (1891-1972), in two performances of “Nobody Knows the Trouble I’ve Seen” from Shawn’s *Four Dances Based on American Folk Music* (1931).



Left: prevalences of François Delsarte’s nine hand gesture archetypes, quantified as the average cosine similarity of all detected hand poses to each archetype, in “Nobody Knows” (2016). Right: extracted frames from the times indicated in the time-series plots above; the left is from Shawn’s own performance in 1938, and the right is from Davon Rainey’s performance in 2016, directed by Adam H. Weinert.



Top: the MIME interface displaying the results of an action similarity search across 31 recorded performances, using 60-element action embedding vectors for each frame of each tracked pose, which are distilled into three actions from the Atomic Visual Actions (AVA) ontology.

Bottom: the results of a pose similarity search across the 31 recorded performances, using 16-element view-invariant pose embedding vectors for each tracked pose, indexed for approximate nearest neighbor search based on cosine similarity.

A UMAP projection of view-invariant pose embeddings (16-element vectors) of a 1% sample of 3 million total key poses detected in the 31 performance videos directed by three contemporary theater directors listed at right. The average embeddings of each director’s entire sub-corpus are represented by large hexagons, and the average embeddings of each work are represented by diamonds.

Krzysztof Warlikowski

- Bluebeard’s Castle (Bartók) / La voix humaine (Poulenc)
- Die Gezeichneten (Schreker)
- Elektra (Strauss)
- Iphigénie en Tauride (Gluck)
- Lady Macbeth of Mtsensk (Shostakovich), pts. 1 & 2
- Les Contes d’Hoffmann (Offenbach)
- Médée (Cherubini)
- The French
- Wozzeck (Berg)

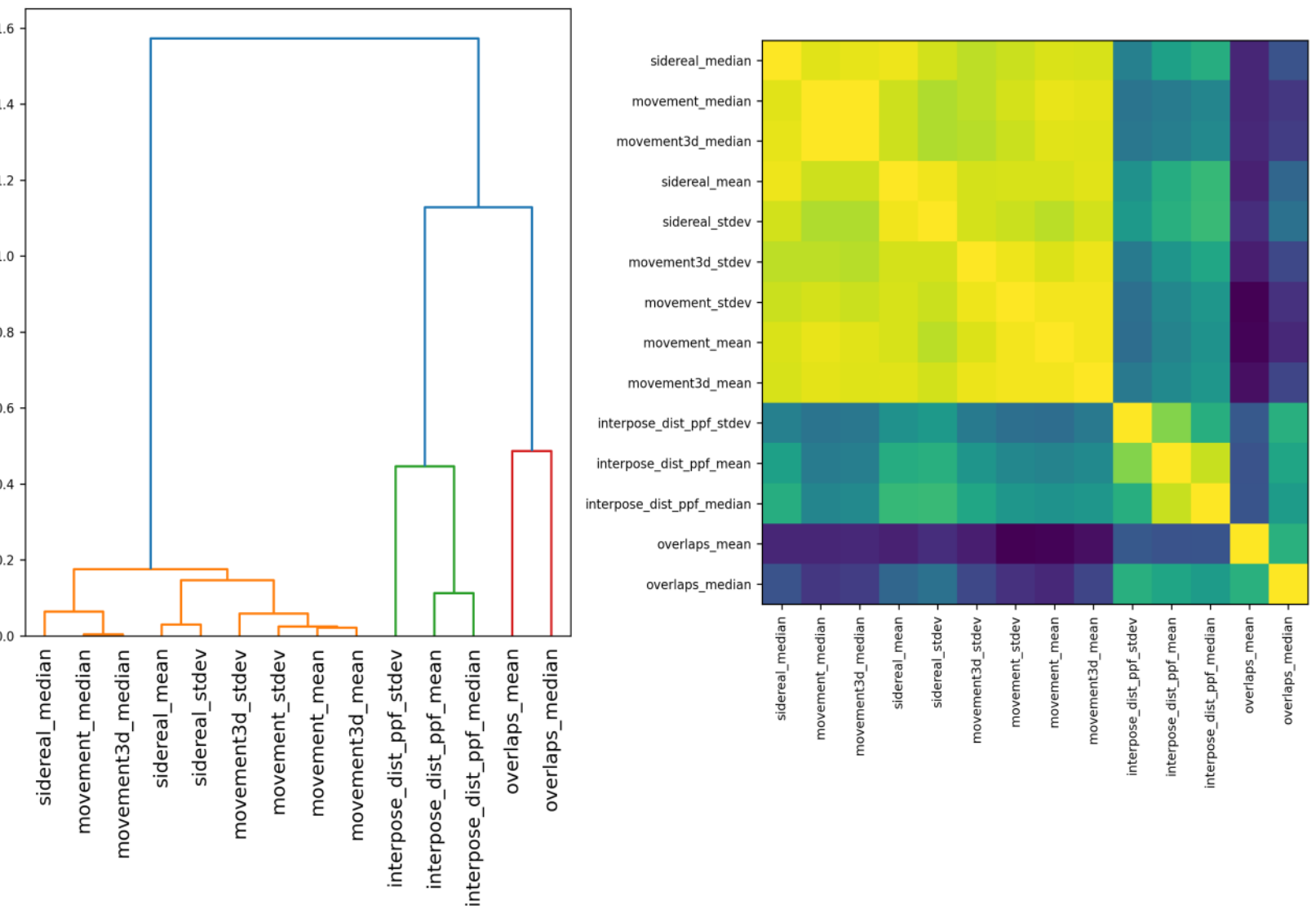
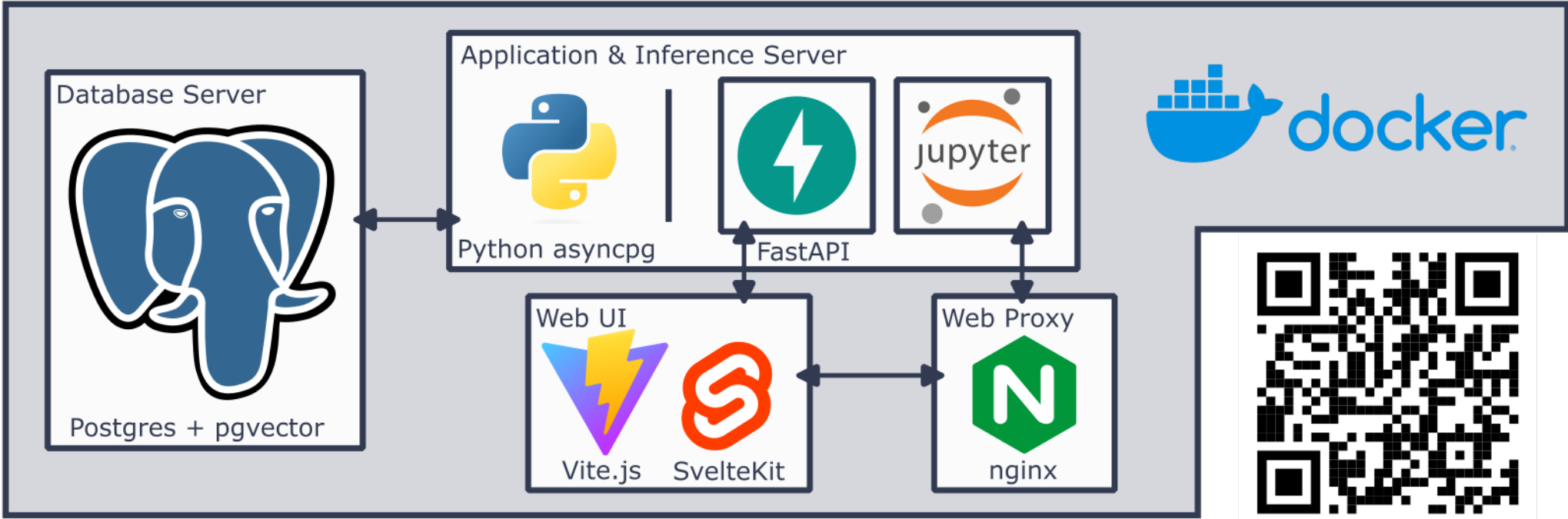
Romeo Castellucci

- Das Floß der Medusa (Henze)
- Democracy in America
- Don Giovanni (Mozart)
- Go Down, Moses
- Inferno
- Parsifal (Wagner)
- Purgatorio
- Requiem (Mozart)
- Resurrección (Gustav Mahler)
- The Magic Flute (Mozart)

Bill T. Jones

- A Letter to My Nephew
- Analogy/Dora: Tramontane
- A Quarrelling Pair
- D-Man in the Waters
- Fondly Do We Hope...
- Fervently Do We Pray
- Holzer Duet... Truisms
- Play and Play
- Secret Pastures
- Story/Time: The Life of an Idea
- We Shall Not Be Moved
- Analogy/Ambros: The Emigrant

MIME system technical architecture



Feature collinearity among pose motion and distance features as derived from a Random Forest classifier trained to differentiate performances directed by one of three directors. The features comprise descriptive statistics for sidereal (relative to the background) and in-place motion based on 2D and 3D armature coordinates, as well as the average number of 3D bounding-box overlaps and the average distances between pose centroids (3D, in meters) in individual frames.

The results of the clustering analysis (left) suggest that the distance and overlap features, at least, might be collapsed to a single most useful feature in further studies.

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